

- Q.27 Explain two arithmetic instructions of 8051 with example.
- Q.28 What are various interrupts of 8051 microcontroller?
- Q.29 Explain serial port operation.
- Q.30 What is maskable and non maskable interrupts?
- Q.31 Explain program status word register.
- Q.32 Write four feature of 8051
- Q.33 Give brief idea of C and Embedded C language.
- Q.34 Write and ALP for adding two 8-bit numbers.
- Q.35 Write steps to execute a stored programme in 8085.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Draw functional block diagram of 8085 and explain function of each block.
- Q.37 What are different addressing modes in 8051? Explain each mode with example.
- Q.38 Write short note (Any one)
- Interfacing of Relay
 - Timer operation
 - Interfacing of 7 segment display

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4th Sem / Mechatronics Subject:- Embedded Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not true about 8085 microprocessor?
- It is an 8-bit microprocessor
 - It is a 40 pin DIP chip
 - It is manufactured using PMOS technology
 - It has 16 address lines
- Q.2 Which of the following is a non-vectored input?
- TRAP
 - RST-7.5
 - RST-6.5
 - INTR
- Q.3 Which of the following is a software interrupt?
- TRAP
 - INTR
 - RST-6.5
 - RST-5

Q.4 How many address lines are required to connect at 4 KB RAM to a microprocessor?

- a) 10 b) 16
- c) 12 d) 20

Q.5 8051 series has how many 16 bit registers?

- a) 2 b) 3
- c) 1 d) 0

Q.6 Read-only memory holds _____ kb of storage in 8051.

- a) 2 b) 3
- c) 4 d) 8

Q.7 Intel 8085 is a _____ bit microprocessor.

- a) 4 bit b) 8 bit
- c) 16 bit d) 32 bit

Q.8 The 8051 microcontroller has _____ I/O pins.

- a) 16 b) 32
- c) 48 d) 64

Q.9 The size of Internal RAM of 8051 is

- a) 128 bytes b) 256 bytes
- c) 64 bytes d) 8 bytes

Q.10 The size of data bus in 8051 microcontroller is _____ bits.

- a) 16 b) 32
- c) 8 d) 64

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define microprocessor.

Q.12 Define data bus.

Q.13 What is assembly language?

Q.14 Write full form of SBUF.

Q.15 Which interrupt in 8085 has least priority?

Q.16 Write full form of SFR.

Q.17 Write full form of PSW.

Q.18 Give one example of data transfer instruction.

Q.19 Write full form of LCD.

Q.20 Define stack.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Draw block diagram of microcomputer and Explain.

Q.22 Compare microcontroller and microprocessor.

Q.23 Explain de-multiplexing of address/data bus in 8085.

Q.24 Draw pin diagram of 8051 microcontroller.

Q.25 Give brief idea of machine language.

Q.26 Explain instruction format of 8085.